

Waterfall Method of Project Management

The Waterfall Method **is a** traditional, sequential project management approach in which a project is divided into clearly defined phases. One phase must generally be completed before the next phase begins.

It is called “Waterfall” because progress flows downward through the stages, like a waterfall.

Basic Structure

Requirements → Planning/Design → Development → Testing → Implementation → Maintenance

Phase	Main Activity	Output
1. Requirements	Identify project requirements	Requirement document
2. Design	Develop technical/project design	Design specification
3. Development	Build the product/system	Working product
4. Testing	Identify and correct defects	Tested product
5. Implementation	Deliver/deploy the product	Operational system
6. Maintenance	Fix problems and make necessary improvements	Maintained system

The Waterfall model follows a sequential approach:

Requirements → Design → Development → Testing → Implementation → Maintenance

In numerical questions, the focus is often on project duration, cost, resource allocation, and phase completion.

Question 1: Project Duration

A software project follows the Waterfall model with the following phases:

Phase	Duration
Requirements Analysis	10 days
System Design	15 days
Development	30 days
Testing	12 days
Implementation	8 days
Maintenance Setup	5 days

Calculate the total project duration.

Answer

Since Waterfall phases are completed **sequentially**:

$$T=10+15+30+12+8+5 \quad T=80 \text{ days}$$

Answer: Total project duration = 80 days.

If the project starts on 1 September, ignoring holidays, the project requires approximately 80 calendar days.

Question 2: Waterfall Project Cost

A company is developing an ERP system using the Waterfall method. The estimated cost of each phase is:

Phase	Cost ()
Requirements	1,20,000
Design	1,80,000
Development	5,50,000
Testing	2,00,000
Implementation	1,50,000
Maintenance	1,00,000

The company also keeps a **10% contingency reserve** on the total estimated cost.

Calculate:

1. Total estimated project cost
2. Contingency reserve
3. Total project budget

Answer

Step 1: Total estimated cost

$$C=1,20,000+1,80,000+5,50,000+2,00,000+1,50,000+1,00,000 \quad C=12,00,000$$

Step 2: Contingency reserve

$$\text{Contingency}=10\% \times 12,00,000 = 1,20,000$$

Step 3: Total project budget

$$\text{Budget}=12,00,000+1,20,000 = 13,20,000$$

Answer:

- Estimated cost = 12,00,000
- Contingency = 1,20,000
- Total budget = 13,20,000

Question 3: Waterfall Schedule and Delay

A project is planned according to the Waterfall model:

Phase	Planned Duration
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Requirements	2 weeks
Design	3 weeks
Development	6 weeks
Testing	3 weeks
Deployment	2 weeks

During testing, a major defect is discovered. Testing takes 2 additional weeks.

The project starts on 5 January.

Calculate:

1. Original project duration
2. Revised project duration
3. Percentage increase in project duration

Answer

Step 1: Original duration

$$T=2+3+6+3+2 \quad T=16 \text{ weeks}$$

Therefore,

Original duration = 16 weeks

Step 2: Revised testing duration

Original testing duration:

3 weeks

Additional delay:

2 weeks

Therefore:

$$\text{Testing}=3+2=5 \text{ weeks}$$

Revised project duration:

$$TR=2+3+6+5+2 \quad TR=18 \text{ weeks}$$

Revised duration = 18 weeks

Step 3: Percentage increase

$$\% \text{ Increase}=(\text{Revised}-\text{Original})/\text{Original} \times 100 =18-16 \times 100 =12.5\%$$

Final Answer

Measure	Result
Original duration	16 weeks
Additional delay	2 weeks
Revised duration	18 weeks
Increase	12.5%

